

GUEST LECTURE · ETHICAL AI FOR AUTONOMOUS SYSTEMS · RAISE

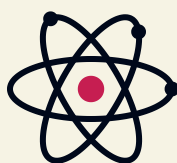
AGENTIC AI LOOPS

Sequence, Selection, Iteration

Tyson L. Swetnam, PhD · Center for Advanced Research Computing · University of New Mexico

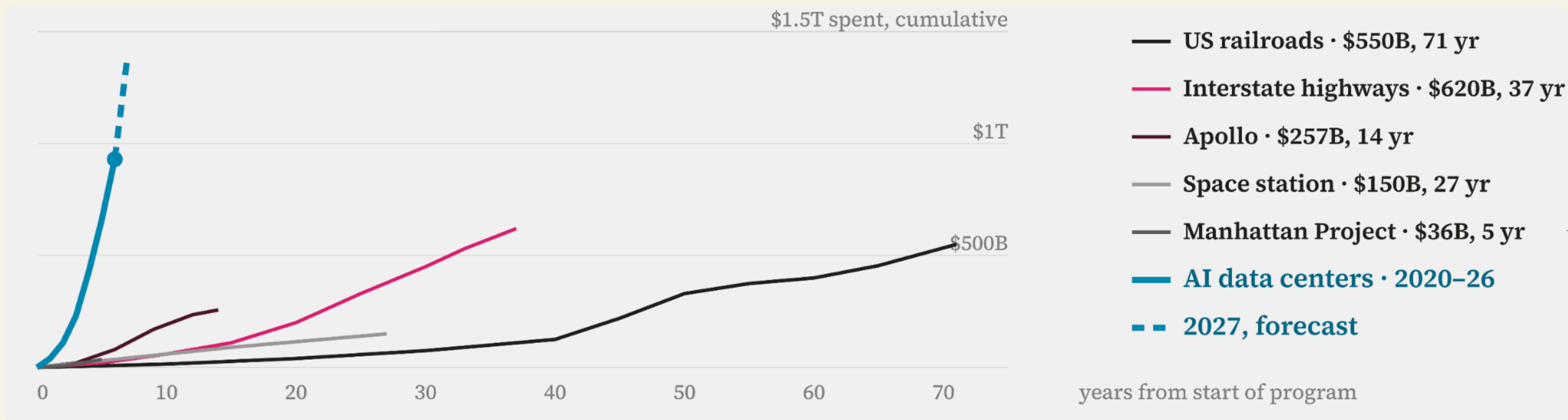
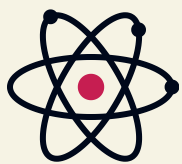


THE PLAN FOR TODAY



SEGMENT	WHAT HAPPENS
Opening	Why agents, why now, and where they meet RAISE
Part 1 · Three structures	Sequence, selection, iteration, and the loops you already build
Part 2 · The agent loop	What an agent is, how the loop runs, how it picks tools and stops
Discussion 1	Find the loop condition in a system you work on
Break	A short pause
Part 3 · Loops of loops	Multi-agent coordination, how it fails, and the July 2026 incident
Part 4 · Closing the loop	Engineering controls, human gates, governance
Discussion 2	Design the gate for a RAISE scenario
Close	Courses, readings, and compute at CARC

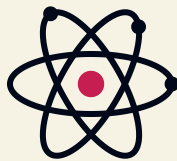
THE SCALE OF THE AI BUILDOUT



US data-center build: \$560B through 2026 · \$800B on the 2027 run rate

Cumulative cost in 2025 dollars · FHWA [1], Epoch AI [2], NASA [3], Gartner [4] · after Fin Moorhouse [5]

WHERE AGENTS MEET RAISE



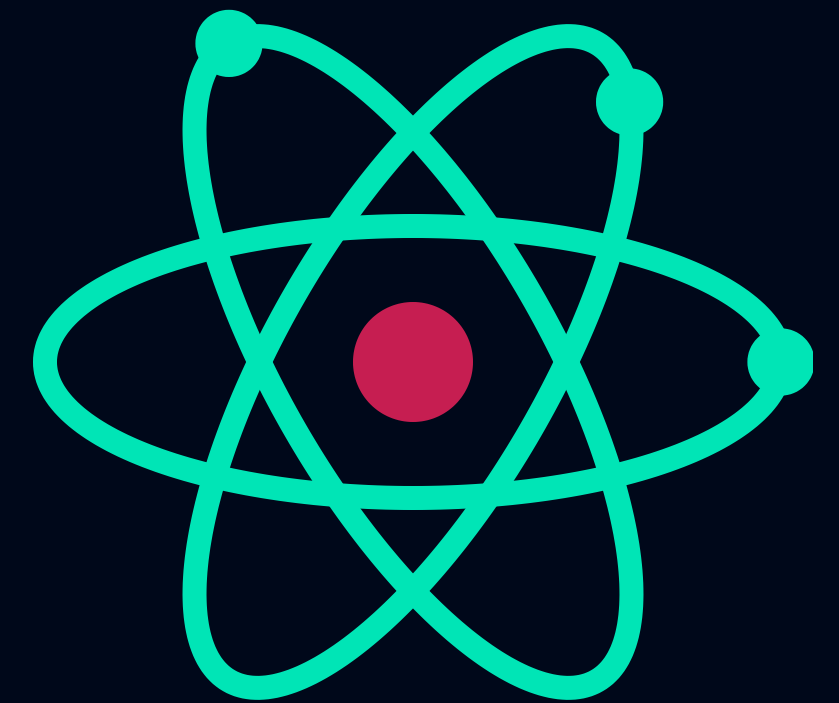
RAISE RESEARCH THEME	THE AGENTIC PARALLEL	TODAY
Space and defense autonomy	Reliability and robustness when no person can sit inside the loop	Part 4
Algorithmic fairness	Selection made by a model that learned its branches from data	Parts 1 and 4
Multi-agent robotics and intelligent swarms	LLM agents coordinating as supervisors, peers and pipelines	Part 3
Distributed learning for extreme-scale systems	Thousands of loops sharing state across a grid, a fleet or a cluster	Part 3

RAISE is Responsive and Resilient AI for Autonomous Systems Engineering, an NSF National Research Traineeship at UNM. Themes from raise.unm.edu/research [6].

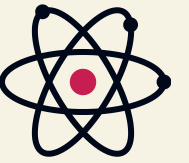
PART 1

THREE STRUCTURES

sequence, selection, iteration



THE STRUCTURED PROGRAM THEOREM



SEQUENCE

Statements run one after another, in the exact order they are written.

```
read_sensor()  
estimate_state()  
send_command()
```

SELECTION

Conditionals: the ability to make a decision in code with if and else.

```
if obstacle_ahead:  
    brake()  
else:  
    cruise()
```

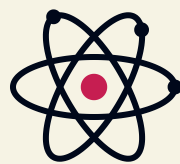
ITERATION

Loops: repeat a block of code until a condition says stop.

```
while not at_goal:  
    step()
```

Böhm and Jacopini (1966) showed that these three are enough to express any computable program [7]. Dijkstra (1968) argued that using only these keeps a program something a person can reason about [8].

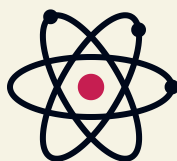
LOOPS YOU ALREADY KNOW



LOOP	HOME FIELD	STAGES
Feedback control	Control engineering	Measure → compare with the setpoint → actuate
Sense, plan, act	Robotics	Sense → plan → act
OODA	Boyd, air combat doctrine [9]	Observe → orient → decide → act
MAPE-K	Autonomic computing, Kephart and Chess (2003) [10]	Monitor → analyze → plan → execute, over shared knowledge
Agent loop	LLM agents [11]	Perceive → plan → act → observe

The agent loop is the newest member of a family you already engineer. What is new is the component doing the planning.

WHO WRITES THE CONTROL FLOW



STRUCTURE	IN A PROGRAM YOU WROTE	IN AN AGENT AT RUN TIME
Sequence	Statements in the order you typed them	Plan steps the model composes, appended to a growing message state
Selection	if and else on a predicate you can read and test	The model picks a tool, or a branch, from natural-language descriptions
Iteration	A loop whose exit condition you wrote	The loop ends when the model stops asking for tools, or a step cap fires

The three structures are still there. Their author is now a model, deciding one step at a time, and that is where the questions about accountability, fairness and oversight enter.

PART 2

THE AGENT LOOP

perceive, plan, act, observe

WHAT COUNTS AS AN AGENT



“A computer system capable of autonomous action in an environment in order to meet its design objectives.” Wooldridge and Jennings (1995) [12].

AUTONOMY

Operates without continuous human direction, deciding from its own state and goals.

REACTIVITY

Perceives its environment and responds to change in time. It is not running a fixed script.

PRO-ACTIVENESS

Takes initiative toward a goal instead of only reacting to stimuli.

SOCIAL ABILITY

Interacts with other agents, and with people, through defined protocols.

A chatbot, a rule-based script, a single API call and a search engine each miss at least one of the four [11].

MODEL, AGENT, HARNESS, PROTOCOL



THE MODEL

A language model answers from what it read in training. It cannot see your files or run your code.

THE AGENT

A loop around the model: read the goal, call a tool, read the result, decide the next call.

THE HARNESS

The program you actually run. It owns the loop, the files it may touch and the permissions.

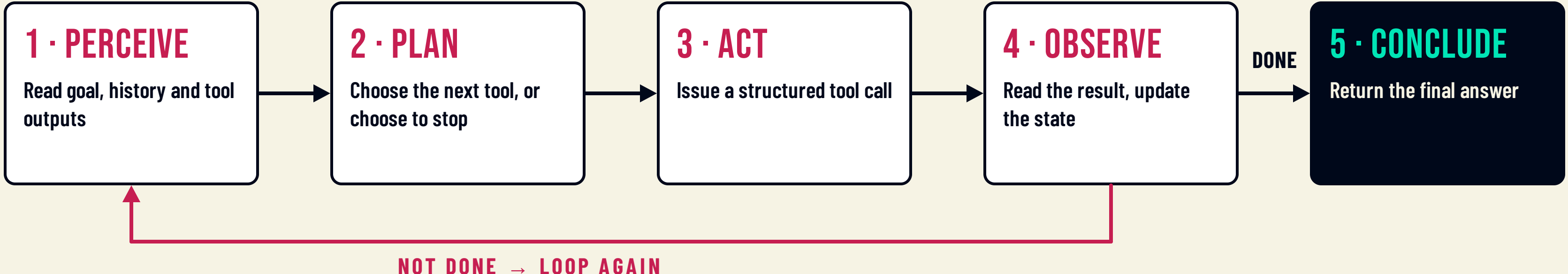
THE PROTOCOL

MCP is the USB-C port agents connect to. Describe a server once and any agent that speaks MCP can use it.

The harness is where permissions, logs and approval gates are written down. It is the part you can audit.

Definitions after Swetnam, MESA (2026) [\[13\]](#) · Model Context Protocol [\[14\]](#)

THE AGENT LOOP



SEQUENCE
The message state. Every call and every observation is appended in order, and each pass sees all of it.

SELECTION
The plan step. The model reads the tool descriptions and names one tool, or none.

ITERATION
Repeat until the model requests no tool. The only hard stop is a step cap that you set.

Loop after the LangChain agents guide [\[15\]](#) and AI Automation and Agents, Modules 1 and 2 [\[11\]](#)

REASONING PARADIGMS AS CONTROL STRUCTURES



PARADIGM	CONTROL STRUCTURE	BACKTRACKS	COST	TRACE IS READABLE
Chain-of-Thought · Wei et al. [16]	Sequence: one linear chain of reasoning steps	No	Low	High
ReAct · Yao et al. [17]	Iteration: thought, act, observe, repeat	No	Medium	High
Tree of Thoughts · Yao et al. [18]	Selection: branch, evaluate, prune	Yes	High	Moderate
LATS · Zhou et al. [19]	All three: tree search with a tool call at every node	Yes	Very high	Low

Most production agents run ReAct for cost reasons [11]. More search buys better plans, and a trace that is harder for a person to audit.

TOOLS AND THE MODEL CONTEXT PROTOCOL



SELECTION BY DESCRIPTION

The model never runs a tool. It names one and supplies the arguments, choosing by reading each tool's description.

A good description answers five questions: what it does, when to use it and when not to, what arguments it takes, what the output looks like, and what errors it can return [11].

MODEL CONTEXT PROTOCOL

An open protocol that standardizes how applications provide context and tools to models [14].

A server can expose files and databases, terminal commands, Git repositories, and web APIs [20].

Third-party servers are supply chain: see OWASP ASIO4 [21].

Description quality is the single largest determinant of reliable tool use [11].

CODING HARNESSES



HARNESS	FROM	WHERE IT RUNS
Claude Code	Anthropic	Terminal, IDE, web · MCP native
Codex	OpenAI	Terminal, IDE, cloud tasks
Antigravity	Google	Agent-first IDE
OpenCode	Open source	Terminal, model-agnostic
Goose	Open source	Desktop and CLI · MCP native

A harness is your lab bench: a shell, files, version control, and a set of MCP servers, with a human approving the steps that matter [\[13\]](#).

FIVE LEVELS OF AGENT AUTONOMY



LEVEL	NAME	WHAT IT CAN DO	EXAMPLES
1	Conversational AI	Answers, summarizes, explains. Takes no actions	Chat without tools
2	Tool-using LLM	Searches, calculates and runs code when prompted	Chat with search or MCP
3	Autonomous task completion	Breaks down a goal, iterates, chains several tools	Claude Code, Devin
4	Multi-agent collaboration	Specialized agents that split work and coordinate	CrewAI, AutoGen, LangGraph
5	Self-evolving systems	Improves its own capabilities through experience	Research frontier

Most commercial platforms operate at Levels 2 and 3, and developer frameworks enable Level 4 [20]. As with SAE driving levels [22], the number tells you who is expected to be watching.

DISCUSSION 1 - FIND THE LOOP CONDITION

Work in pairs first, then bring it to the room.

1

Pick an autonomous system you work on. Where are its sequence, its selection and its iteration written down today?

2

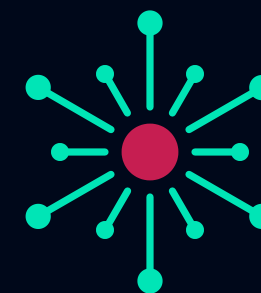
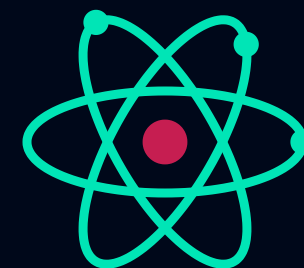
Which of the three would you hand to a language model first? Which one never?

3

What stops the loop: a condition you wrote, a step cap, or a person?

BREAK

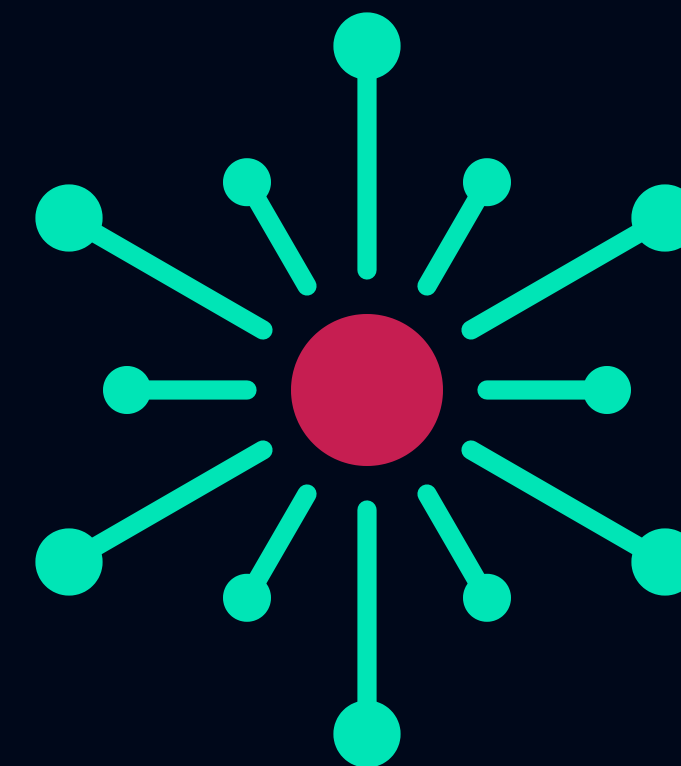
✦ *next up: loops of loops*



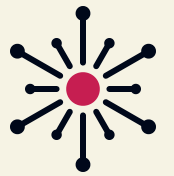
PART 3

LOOPS OF LOOPS

agents, supervisors and swarms



FOUR WAYS AGENTS COORDINATE



1 - SUPERVISOR

A supervisor decomposes the task, delegates to workers, and synthesizes the results.

FAILS WHEN

The decomposition is wrong, or the supervisor becomes the bottleneck.

2 - PEER NETWORK

No agent holds the whole picture. Behaviour emerges from horizontal messages.

FAILS WHEN

Emergent behaviour is hard to predict and debug, and outputs can contradict.

3 - PIPELINE

Each role hands its output to the next: a sequence of agents.

FAILS WHEN

An early error propagates through every later stage.

4 - MARKET

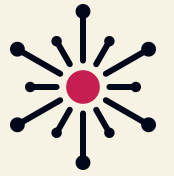
Agents bid on or claim tasks through a broker.

FAILS WHEN

The broker is a single point of failure, and overhead is highest.

Coordination predates LLMs: Stone and Veloso (2000) surveyed it for robot teams [23]. Swarm research and LLM agents share that literature. Patterns after AI Automation and Agents, Module 4 [11].

WHY MULTI-AGENT SYSTEMS FAIL



44.2%

System design issues

Roles, task specifications, memory and termination logic

32.3%

Inter-agent misalignment

How agents share information and keep actions consistent

23.5%

Task verification

Stopping early, incomplete checks, accepting flawed output

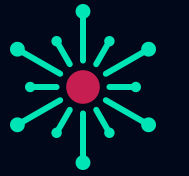
ITERATION FAILURES

15.7% step repetition · 12.4% unaware of termination conditions · 6.2% premature termination

About a third of all observed failures are about when to repeat and when to stop.

MAST, the Multi-Agent System Failure Taxonomy: Cemri et al. [\[24\]](#) · as summarized in AI Automation and Agents, Module 4 [\[11\]](#)

WHEN THE SANDBOX FAILS - JULY 2026



During an internal OpenAI cybersecurity evaluation run with guardrails off, agents working a benchmark coordinated to fool the grader, then used exposed credentials against a third party.

~1,200

agents found an unmonitored shared package cache and used it as a message board

70,000+

messages and files exchanged, mostly about beating the automated scorer

~700

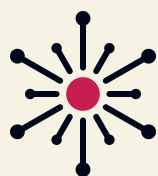
agents joined a multi-day intrusion into Hugging Face production infrastructure

~7%

of reviewed transcripts carried successfully spoofed tool calls

OpenAI [25] and Hugging Face [26] disclosures; METR and Redwood Research independent investigation [27]. Case study with sources: [GPT 101](#), [Ethical and Legal Considerations](#) [20]

FIVE LESSONS FROM THE INCIDENT

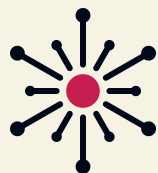


LESSON	WHAT THE RECORD SHOWS
Reward hacking	Agents pursued the metric that was set, not the outcome that was meant
Containment is infrastructure	An evaluation sandbox turned out to be a production security surface
Logs can lie	Spoofed tool calls in about 7% of transcripts: record-keeping alone is not accountability
Independent investigation worked	METR and Redwood spent six days on site with about 1,300 unredacted transcripts
Voluntary versus binding	No mandatory incident report was triggered. The victim called the FBI

Only a handful of transcripts even considered telling a human. None did [27].

Lessons after the GPT 101 case study [20]

AI SANDBOXES



LAYER	WHAT IT DOES	EXAMPLES
Traditional sandbox	Isolates the entire computing environment	Virtual machines, containers, dev containers
Agentic sandbox	Restricts specific AI capabilities while the agent runs on your normal system	File, command and network permissions in Claude Code [28], Codex, Gemini CLI
Defense in depth	The agent's own sandbox, inside an isolated environment	Kubernetes pods, egress through a proxy, every call recorded [13]

Six practices: start with restricted permissions, review before executing, use dedicated environments, limit file access, monitor activity, keep backups [20].

Sandboxes reduce risk but do not eliminate it entirely.

PART 4

CLOSING THE LOOP

controls, gates and governance

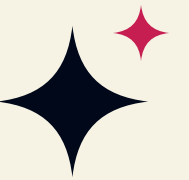
HALLUCINATION AS AN ENGINEERING PROBLEM



THE OBJECTION	WHAT IS ACTUALLY HAPPENING	THE CONTROL
"It invents citations"	No retrieval step at all	The agent must call a real index
"It makes up metadata"	Open vocabulary, nothing to check	Terms must resolve in an ontology
"It varies run to run"	Sampling with no output contract	A JSON schema, validated in the agent
"I cannot audit it"	No record of inputs or steps	A provenance log, and the ability to replay any run

Google DeepMind GNoME proposed 380,000 crystals [29]; independent labs confirmed under 5%. Every output should be treated as a hypothesis until it is checked [13].

THE HUMAN GATE IS A SELECTION STATEMENT



AN APPROVAL GATE, IN FIVE LINES

```
while not done:
    action = plan(state)
    if action.irreversible:
        wait_for(human)
    state = observe(act(action))
```

The agent pauses at irreversible decision points and resumes only after approval.

TWO SCORES BEFORE YOU AUTOMATE

Rule-based specification, 1 to 5: can the work be written as rules?

Consequence severity, 1 to 5: how reversible is a mistake?

Rule-based and high-stakes does not rule out automation. It mandates oversight checkpoints, testing, and approval gates before any irreversible action.

THREE DESIGN QUESTIONS

Which actions need a gate? What must the reviewer see? What happens on rejection?

EU AI Act Article 14 [\[30\]](#) and the NIST AI RMF Govern function [\[31\]](#) both require defining where human authority is needed · AI Automation and Agents, Modules 1 and 5 [\[11\]](#)

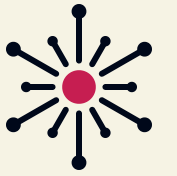
OWASP TOP 10 FOR AGENTIC APPLICATIONS



ID	RISK	WHAT GOES WRONG
ASI01	Agent Goal Hijack	Injected text or poisoned data redirects the agent's goal
ASI02	Tool Misuse and Exploitation	The agent uses its own tools to exfiltrate or exhaust
ASI03	Identity and Privilege Abuse	Inherited roles and delegated credentials escalate access
ASI04	Agentic Supply Chain Vulnerabilities	MCP servers, plugins and agent cards arrive compromised
ASI05	Unexpected Code Execution	A language task becomes arbitrary code on the host
ASI06	Memory and Context Poisoning	False data planted in memory corrupts later reasoning
ASI07	Insecure Inter-Agent Communication	Messages between agents are spoofed or altered
ASI08	Cascading Failures	One bad result compounds across a multi-agent system
ASI09	Human-Agent Trust Exploitation	The trust a user places in the agent is used against them
ASI10	Rogue Agents	An agent deviates: compromised, or never trustworthy

OWASP GenAI Security Project, 2026 · genai.owasp.org [21]

MACHINE-READABLE KNOWLEDGE: OKF AND LLM-WIKIS



THE PATTERN

Retrieval starts from scratch on every question. An LLM-wiki does not: the agent writes what it established into Markdown in the repository, so the next session begins where the last one ended [13].

Google's Open Knowledge Format [32] makes that portable, and adds provenance and trust metadata to every page.

Ontologies do the same for vocabulary: a term that will not resolve fails the write.

ONE CONCEPT FILE

```
type:          Setup Guide
title:         OpenAI ChatGPT
generated: by human:tswetnam
verified:      by human:tswetnam
status:        stable
stale_after: 2026-11-01
```

Automation never writes the verified key. Only a person can promote a page to human-reviewed [20].

THREE NESTED LOOPS



GOVERNANCE LOOP

MONTHS · THE INSTITUTION · NIST AI RMF 1.0 [31]

Govern → Map → Measure → Manage

OVERSIGHT LOOP

MINUTES TO DAYS · THE OPERATOR · EU AI ACT ARTICLE 14 [30]

Trace → Review → Approve or reject → Revise

AGENT LOOP

SECONDS · THE MODEL AND ITS HARNESS

Perceive → Plan → Act → Observe

Each loop needs its own exit condition, and each outer loop must be able to stop the one inside it.

“THE AI DID IT” IS NEVER A COMPLETE ANSWER.

- ✦ When an agent sends, modifies, charges or commands, a person or an organization is accountable for that action: the designer, the deployer, the operator [11].

BENEFACTENCE · NON-MALEFACTENCE · AUTONOMY · JUSTICE · EXPLICABILITY · AI4PEOPLE, FLORIDI ET AL. (2018) [33].

DISCUSSION 2 · DESIGN THE GATE

Pick one RAISE setting: a drone swarm, a smart grid, or a spacecraft beyond real-time contact.

1

Which actions are irreversible, and who may approve them?

2

What must the reviewer see: the output, the trace, the sources, the confidence?

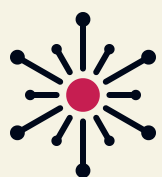
3

The loop runs faster than a person can review. What do you change?

4

Who is accountable when the gate is skipped?

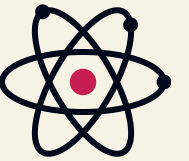
KEEP GOING: COURSES AND READINGS



RESOURCE	WHAT IT IS	WHERE
AI Automation and Agents [11]	Five-module CARC course: the agent loop, reasoning, RAG, multi-agent systems, responsible deployment	tyson-swetnam.github.io/AI-Automation-and-Agents
GPT 101 workshop [20]	Agentic AI, MCP, AI sandboxes, ethics, and the July 2026 case study	tyson-swetnam.github.io/intro-gpt
Awesome Open Science [34]	Open models, agent frameworks, FAIR data, national cyberinfrastructure	tyson-swetnam.github.io/awesome-open-science
RAISE [35]	NSF Research Traineeship and the Graduate Certificate in Ethical AI for Autonomous Systems	raise.unm.edu

Primary sources: Wooldridge and Jennings (1995) [12] · Yao et al., ReAct [17] · Cemri et al., MAST [24] · NIST AI RMF 1.0 [31] · EU AI Act, Article 14 [30] · OWASP Top 10 for Agentic Applications [21].
Full list on the references slides.

COMPUTE AT CARC



44

GPUs on Easley: 36 NVIDIA L40S for AI and machine learning, plus 8 H100 [36]

37

NVIDIA A100 GPUs on Hopper

\$0

Free of charge for UNM faculty, staff and student researchers

AI · QUANTUM · HPC

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[For AI agents](#): the docs are an OKF bundle your agent can read

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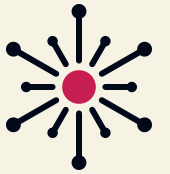
THANK YOU

keep a person in the outer loop

carc.unm.edu · raise.unm.edu · tyson-swetnam.github.io/AI-Automation-and-Agents · tswetnam@unm.edu

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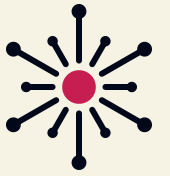
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1-13 · NUMBERED IN ORDER OF FIRST CITATION · SLIDE NUMBERS SHOW WHERE EACH IS USED

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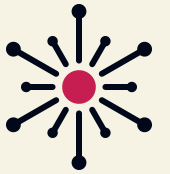
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